

ALEXANDER ARUTCHEV

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EDUCATION

Glenbrook North High School 2023 – Present

Northbrook, IL · GPA: 4.96/5.00 Weighted, 3.97/4.00 Unweighted

- Glenbrook Academy Program; one of 14 students selected in grade
- Coursework: AP Calculus BC (5), AP Computer Science A (5)
- Currently Taking: AP Statistics, AP Biology; Multivariable Calculus & Linear Algebra (dual credit, Oakton Community College)
- Activities: Math Team (2× Harvard-MIT Math Tournament qualifier, one of 6 from GBN), Varsity Chess Team, Varsity Tennis (one of 3 singles players)

PUBLICATION

- Pranav Anuraag, Ethan Xu, Alexander Arutchev, Asher Nerenberg. “*SENTINEL: Sentiment Evolution and Narrative Tracking in Extended LLM Interactions.*” **MTI-LLM Workshop @ NeurIPS 2025.**

WORK EXPERIENCE

AI Researcher · Algoverse AI Research June – December 2025

Remote

- Co-designed novel empirical study on sentiment drift across five frontier LLMs (GPT-4.1, Claude-Sonnet-4, DeepSeek-R1, Qwen3, Llama-4-Maverick) in multi-turn agentic workflows
- Developed three quantitative metrics including a novel Drift Propagation Index (DPI); validated results via one-way ANOVA (η^2p up to .20, $p < .001$)
- Helped build token-level Integrated Gradients attribution pipeline with Captum; found 12 tokens account for 73.8% of sentiment-bearing signal across 3,092 highlighted tokens
- Sourced, cleaned, and processed three datasets (IMDB reviews, MultiFC news, Big Five essays); full experimental pipeline in PyTorch, Pandas, NumPy, and Matplotlib

Math Team Coach · Northbrook Junior High School September 2024 – Present

Northbrook, IL · Volunteer

- Designed weekly lesson plans for 10–20 students covering number theory, combinatorics, algebra, and geometry for MathCounts preparation
- Built an interactive web platform with games and slides (HTML/CSS/JavaScript) to supplement in-person lessons
- Coached the first student from NBHJ to qualify for MathCounts State in five years

AWARDS & TEST SCORES

- ACT: 36/36 · SAT: 1550/1600 · PSAT/NMSQT: 1520/1520
- AIME Qualifier
- AMC10: 97.5 · AMC12: 91.5
- Illinois Math State: 2nd Place Individual
- USACO: Silver Division
- AP Scholar

SKILLS

Programming: Python, Java, JavaScript / HTML / CSS

ML & Data Science: PyTorch, spaCy, Captum, Pandas, NumPy, Matplotlib

Tools: GitHub, Jupyter Notebook, Overleaf